The operation of finding the direct connection between x and y without the presence of the auxiliary variable t is called **elimination of the parameter**. The question as to why should we deal with parametric equations is that two or more variables are reduced to a single variable, t.

For example, the equation of a circle with centre (0, 0) and radius r is $x^2 + y^2 = r^2$ This equation describes the relationship between x and y and the equations

$x = r\cos t$; $y = r\sin t$ are parametric equations.

Conversely, if we eliminate t, we get $x^2 + y^2 = r^2$

If y is regarded as a function of x then $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)}$.

If we regard x as a function of y, then the derivative of x with respect to y is,

$$\frac{dx}{dy} = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{f'(t)}{g'(t)}.$$

In the case of the circle, then $\frac{dy}{dx}$ is the slope of the tangent to the circle namely

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{r\cos t}{-r\sin t} = -\cot t \ .$$

Example 10.26

Find $\frac{dy}{dx}$ if $x = at^2$; $y = 2at$, $t \neq 0$.

Solution

We have $x = at^2$; $y = 2at$

$$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{2a}{2at} = \frac{1}{t}.$$

Example 10.27

Find $\frac{dy}{dx}$ if $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

Solution

We have $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

Now $\frac{dx}{dt} = a(1 - \cos t)$; $\frac{dy}{dt} = a\sin t$

Therefore, $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{a\sin t}{a(1-\cos t)} = \frac{\sin t}{(1-\cos t)}$.

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